Same Engagement, Different Outcomes

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Data Mining Course Project

Motivation

Online learners can click the same amount and finish differently. We ask which temporal and resource-use patterns distinguish Pass/Distinction from Fail among students with similar first-month engagement. Withdrawn is a separate outcome, not a missing label.

Data and leakage audit

OULAD: 32,593 enrolments, 10.65M VLE rows. Dates are relative to course start and include negative days. The click stream does **not** stop at `date_unregistration`: 1272 withdrawn students have 29440 post-unregistration click rows. We therefore keep the main cohort and treat at-risk filtering as a sensitivity analysis.

Design

W1 = days 0–27. W2 = 28 to min(55, first TMA−1), with a fixed 28–55 fallback when the first TMA is before week 4. Strict features exclude scores, banked flags, final results and post-cutoff clicks. Splits are by student; temporal and LOMO folds are student-disjoint. CEM uses module × presentation exact strata and coarsened W1 clicks, active days and entropy. Clustering uses W1 profiles with 300 bootstrap refits; models M0–M5 form a feature ladder. A specification curve and diagnostics test robustness.

Cohort definitions

T1 = Pass/Distinction vs Fail, Withdrawn excluded (n=22,437; base rate 68.6%). T2 folds Withdrawn into the negative class; T3 keeps three classes. The main cohort is not filtered on `date_unregistration`; at-risk filtering is a sensitivity analysis.

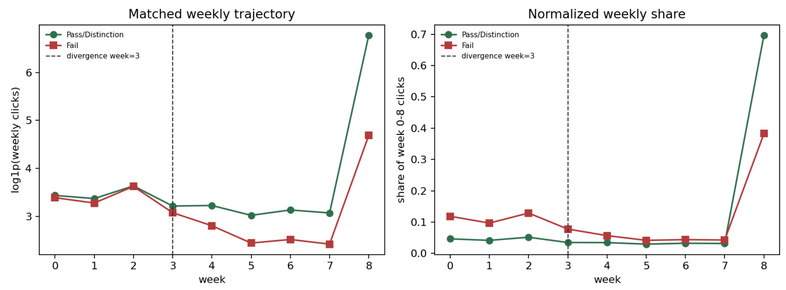
Matching and clustering

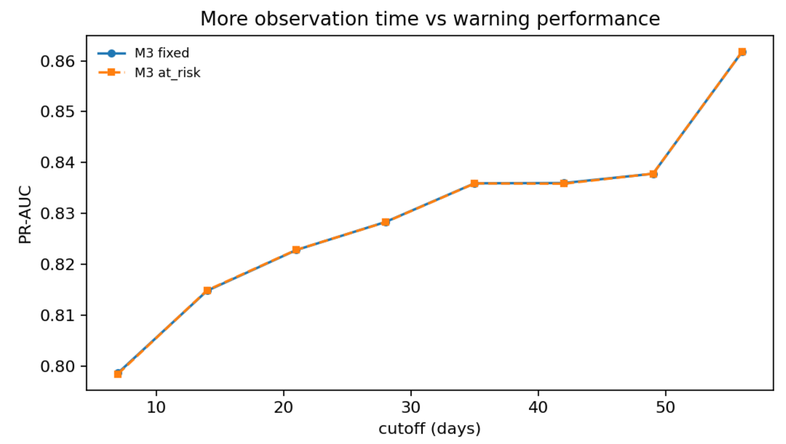
CEM matched 20660 T1 enrolments; coarsened max |SMD| 0.000, raw max |SMD| 0.033, treated support 0.900, E-value 1.555. k-means selected k=2 (bootstrap ARI 0.913, minimum Jaccard 0.810); clusters are descriptive.

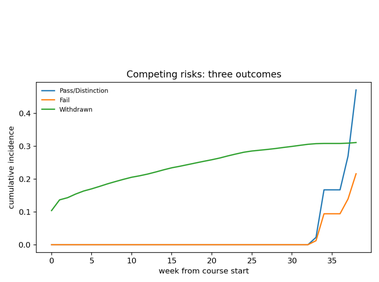
Main results

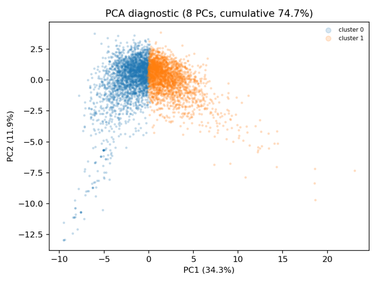
- M1 held-out test: PR-AUC 0.827, ROC-AUC 0.724.
- Grouped CV ROC-AUC 0.719; 2013→2014 0.707; LOMO 0.699.
- M3 strict PR-AUC 0.828; M5 0.838; M4 (scores) is an upper bound.
- Context-only PR-AUC 0.783 (ROC-AUC 0.657; no-skill 0.677); context+volume 0.853; context+M1 0.854.
- Matched trajectory divergence week: 3 (inside W1; early similarity is approximate).
- Random-label control ROC-AUC 0.530, close to chance.
- CEM balance is exact on coarsened bins; raw max |SMD| is 0.033 and treated support is 0.900.

Figures









Diagnostics and placebo checks

Quiz inclusion changes M1/M3 PR-AUC by -0.001/-0.000. Exposure-definition CEM gives W2 click differences from -27.621 to 37.558; seven bin grids range from 34.451 to 39.814. The repeater placebo reaches ROC-AUC 0.657.

Takeaways

- Total clicks are an entry point; distribution and composition add a small, setting-dependent increment.
- Row-level splitting leaks students; grouping changes the answer.
- Withdrawn must be modelled separately.
- Calibration and external validation matter more than a small random-split gain.

Limitations

Observational associations only; measured W1 matching cannot remove unmeasured confounding. Clicks are a weak proxy for study time. Pre-start and future-window diagnostics are not null, so they are not clean leak detectors.

Disclosures

Public OULAD data. No external collaborators. Claude Code (Anthropic) was used for code scaffolding, pipeline execution, figure generation and drafting; the group is responsible for the final submission.

What changes the answer

Pre-start-only PR-AUC 0.782 is a strong signal, not a clean control; the future-window control is 0.861 (a placebo failure, not a leak test). Random-label ROC-AUC is 0.530. Context-only PR-AUC is 0.783 vs no-skill 0.677 (ROC-AUC 0.657), so the context contrast is descriptive.

Reproducibility

Frozen config hash sha256:1719a…; full hash in `config/config.sha256`. Unit tests, staged S00–S12 logs, `STATUS.final.json` and `validate_run.py` output are included in the submission archive.

Bottom line

Total clicks are an entry point, not an explanation. Early-window behaviour adds a small and setting-dependent predictive increment; calibration, external validation and the separate treatment of Withdrawn matter more than a single random-split score. All reported contrasts are associations, not causal effects.

LaTeX TikZposter